

2023 Ch H2 Q3(b)

Section: Nature's Chemistry

Topic: Proteins (Casein)

Question Summary

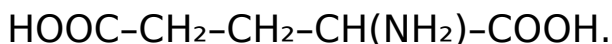
Proteins in cheese — amino acids in casein, essential amino acids, and protein denaturation on heating.

Worked Solution

According to the SQA Marking Instructions for 2023 H2 Q3:

(i) Amino acids that must be obtained from the diet (e.g. leucine, valine) are essential amino acids.

(ii) The structure for glutamic acid (one of the amino acids present):



(iii) Heating proteins disrupts the secondary/tertiary structure (H-bonding, ionic and hydrophobic interactions), causing denaturation.

Final Answer

(i) Essential amino acids.

(ii) Glutamic acid:



(iii) Heating causes denaturation (loss of 3D structure).

Revision Tips

- 'Essential' means cannot be synthesised by the body in sufficient amounts — must be supplied by diet.
- Protein denaturation changes shape → changes function (e.g., texture changes in heated cheese).
- Draw amino acids with both amine and carboxyl groups; remember zwitterion form in neutral solution.